GOD'S EYE: OMNIVISION INTELLIGENCE PLATFORM

An Open Source Intelligence (OSINT) Research Platform with AI-Powered Analysis

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Abstract

This paper presents God's Eye (OMNIVISION), an advanced educational Open Source Intelligence (OSINT) platform developed at Presidency University, Bangalore. The system aggregates data from 45+ publicly available APIs across 12 intelligence modules, providing a unified dashboard for research, cybersecurity training, and digital forensics education. The platform integrates face recognition, cyber threat analysis, geospatial tracking, news sentiment analysis, and AI-powered reporting via LLaMA 3.3 70B through the Groq inference engine. Secured by JWT-based authentication with role-based access control, God's Eye offers a complete pipeline from raw data ingestion to structured intelligence reports. This paper details the system architecture, data processing pipeline, API integration strategy, risk scoring methodology, and ethical framework. The platform demonstrates how freely available data sources, combined with modern full-stack engineering and large language models, can produce a professional-grade OSINT capability for academic and cybersecurity education purposes.

Keywords: OSINT, Open Source Intelligence, Cybersecurity, FastAPI, React, LLaMA, Threat Intelligence, Digital Forensics, API Aggregation, JWT Authentication

1. Introduction

Open Source Intelligence (OSINT) is the practice of collecting and analysing publicly available information to produce actionable intelligence. In the era of big data and interconnected digital systems, the ability to aggregate, correlate, and interpret open-source data is a critical skill in cybersecurity, journalism, law enforcement, and academic research. Traditional OSINT work requires manually navigating dozens of separate tools and platforms, a process that is time-consuming, inconsistent, and difficult to teach systematically.

God's Eye addresses this challenge by providing a single, unified OSINT platform that automates the aggregation of 45+ public data sources into an integrated intelligence dashboard. The platform is designed for use in academic and cybersecurity training environments, providing hands-on exposure to real intelligence workflows while enforcing ethical usage through mandatory ethics agreements and comprehensive activity logging.

This paper makes the following contributions:

- A full-stack OSINT platform architecture combining a React frontend with a Python FastAPI backend
- An asynchronous API aggregation engine capable of querying 45+ external services in parallel
- An AI-powered intelligence reporting system using LLaMA 3.3 70B via the Groq inference API
- A multi-signal threat scoring model combining six independent risk indicators into a unified 0–100 risk score
- A Maltego-style intelligence graph for visualizing entity relationships
- A structured ethical and legal framework governing all platform usage

2. Related Work

OSINT platforms have evolved considerably over the past decade. Tools such as Maltego provide graph-based link analysis for investigators but require paid licensing and offer limited automation. SpiderFoot and Recon-ng are open-source frameworks popular in penetration testing communities; however, they are command-line oriented and lack accessible interfaces for educational use.

Commercial threat intelligence platforms like Recorded Future and ThreatConnect aggregate news and dark web data but are enterprise-focused with substantial cost barriers. Academic work by Miorandi et al. (2012) and Hu et al. (2015) has explored large-scale social data mining, while more recent research has examined the application of large language models to intelligence summarisation tasks.

God’s Eye occupies a distinct space: a fully open-source, API-first OSINT platform with a professional UI, LLM integration, and an explicit educational mission. By building on free API tiers, it eliminates cost barriers while delivering capabilities comparable to commercial tools. The integration of LLaMA 3.3 70B for automated report generation represents a novel application of generative AI to the intelligence analysis workflow.

3. System Architecture

God’s Eye is built on a modern three-tier architecture comprising a React-based frontend, a Python FastAPI backend, and an external API aggregation layer. The system is designed for local deployment on Windows 10/11 (64-bit) with Python 3.10+ and Node.js 18+, and can be launched via a single batch script.

3.1 Frontend Architecture

The frontend is implemented in React 18 with Vite as the build tool, providing fast hot-module replacement during development and optimised production bundles. Tailwind CSS provides utility-first styling supporting a dual-theme system: ‘God’s Eye Mode’ (dark red/black) and ‘Analyst Mode’ (professional blue/grey). Client-side routing is handled by react-router-dom v6. API communication uses Axios, and JWT tokens are managed through the React Context API.

Component	Technology	Purpose
UI Framework	React 18 + Vite	Component-based SPA
Styling	Tailwind CSS + inline styles	Dual-theme OSINT UI
Routing	react-router-dom v6	Client-side navigation
HTTP Client	Axios	Backend API calls
Maps	Leaflet.js + OpenStreetMap	Flight and geo tracking
Graphs	Custom SVG + animations	Intelligence Graph visualisation

3.2 Backend Architecture

The backend is implemented in Python using FastAPI with Uvicorn as the ASGI server, providing asynchronous request handling essential for concurrent API calls. External services are accessed via httpx in async mode. Authentication is handled by PyJWT with bcrypt password hashing. The Groq SDK provides access to LLaMA 3.3 70B for AI-powered analysis and report generation. Image processing for EXIF extraction uses Pillow (PIL), and all API keys are managed through python-dotenv.

Layer	Technology	Purpose
Web Framework	FastAPI + Uvicorn	Async REST API server
HTTP Client	httpx (async)	External API calls
Authentication	PyJWT + bcrypt	JWT token auth
AI Engine	Groq SDK (LLaMA 3.3 70B)	Intelligence reports and chat
Image Processing	Pillow (PIL)	EXIF + GPS extraction
Concurrency	asyncio.gather()	Parallel API calls

3.3 API Endpoints

The backend exposes eight primary route groups corresponding to the major intelligence domains:

- /auth — JWT authentication and role-based access control
- /identity — face recognition, reverse image search, person intelligence
- /cyber — IP tracking, domain investigation, breach detection, VirusTotal scanning
- /geo — live flight tracking, ship tracking, weather, satellite imagery
- /news — news aggregation, sentiment analysis, GDELT global monitoring
- /visual — EXIF metadata extraction, GPS forensics, image analysis
- /ai — LLaMA 3.3 70B chat interface and automated investigation engine
- /recon — username enumeration across 30+ platforms, multi-source breach checking

4. Data Pipeline and Investigation Workflow

The investigation pipeline follows a seven-stage process from user query to final intelligence report. This standardised workflow ensures consistency, reproducibility, and auditability across all OSINT investigations conducted within the platform.

4.1 Pipeline Stages

Stage 1: User Query. The investigator enters a target entity—which may be a person's name, IP address, domain name, email address, or image—through the web interface.

Stage 2: Input Validation and Routing. The frontend and backend sanitise the input to prevent injection attacks and route the request to the appropriate intelligence module based on the input type and selected analysis mode.

Stage 3: Parallel API Calls. Python's asyncio.gather() dispatches all relevant external API requests simultaneously rather than sequentially. This architectural decision reduces total query time from approximately 20 seconds (sequential) to approximately 5 seconds (parallel) for a face scan involving four concurrent APIs.

Stage 4: Data Aggregation. Results from individual APIs are normalised into a common data format, handling inconsistencies in field naming, date formats, and confidence score scales across different providers.

Stage 5: AI Analysis. Aggregated data is passed to LLaMA 3.3 70B via the Groq API, which generates a structured intelligence report in natural language. The LLM prompt is engineered to produce professional analyst-style output with explicit confidence levels and actionable recommendations.

Stage 6: Report Generation and Export. The final intelligence report is presented in the dashboard and can be exported in three formats: PDF (with God's Eye branding and educational watermark), JSON (structured machine-readable data), and plain text.

Stage 7: Activity Logging. Every investigation is logged with the investigator's username, timestamp in IST, target entity, module used, and action taken. Logs are accessible to ADMIN and RESEARCHER roles only.

4.2 Intelligence Correlation Chains

Beyond single-source lookups, God's Eye implements multi-step correlation chains that automatically chain data from one source into queries for the next. The following chains are implemented:

- Email Investigation: LeakCheck breach data → domain extraction → WHOIS lookup → hosting IP → AbuseIPDB reputation check
- Domain Investigation: DNS A record resolution → IP geolocation → AbuseIPDB → VirusTotal → URLScan → AI report
- Person Investigation: Name search → NewsAPI mentions → Reddit posts → AI OSINT summary → identity report
- Image Forensics: EXIF extraction → GPS coordinate extraction → reverse geocoding → location intelligence
- IP Analysis: Geolocation → ISP identification → AbuseIPDB → VirusTotal → Shodan device enumeration → AI analysis

5. Multi-Signal Threat Scoring Model

A key contribution of this work is the Threat Score Calculator, a multi-signal risk aggregation model that combines six independent indicators into a unified risk score on a scale of 0 to 100. The calculation is performed entirely in the browser as a pure frontend computation, enabling instant results without additional API calls.

5.1 Signal Weights

Signal	Max Points	Trigger Condition
VirusTotal malicious engines	30 pts	5 points per malicious engine detected
AbuseIPDB confidence score	25 pts	Score percentage × 0.25
Domain age (new = risky)	15 pts	Under 7 days = 15pts; under 30 days = 10pts
Data breach presence	15 pts	3 points per breach (maximum 5 breaches counted)
GPS in image metadata	10 pts	GPS coordinates detected in EXIF = 10pts
Negative news sentiment	5 pts	News sentiment analysis returns NEGATIVE

5.2 Risk Classification

Score Range	Risk Level	Recommended Action
75–100	CRITICAL	Immediate action required
50–74	HIGH	Investigate further urgently
25–49	MEDIUM	Monitor closely
0–24	LOW	No significant threat indicators

The weighting scheme was designed to reflect the practical significance of each indicator in real-world threat intelligence. VirusTotal's 70+ engine consensus carries the highest weight (30 points) because multi-engine agreement is the strongest available signal for malicious infrastructure. AbuseIPDB's community-sourced abuse score is weighted second (25 points) as it reflects observed active malicious behaviour. Domain age is weighted third (15 points) because newly registered domains are disproportionately associated with phishing and malware campaigns.

6. Intelligence Modules

God's Eye provides 12 distinct intelligence modules, each targeting a specific domain of open-source investigation. The following sections describe the capabilities of the core modules.

6.1 Identity Engine

The Identity Engine supports two investigative modes. In face scan mode, the investigator uploads an image which is simultaneously submitted to four parallel services: Luxand.cloud (age estimation, gender detection, emotion analysis, smile percentage, glasses detection), Face++ (comprehensive facial attribute analysis including beauty scoring and emotion breakdown), SauceNAO (reverse image search with similarity scoring), and automated search links to Yandex Visual Search, Google Lens, and Bing Image Search. EXIF metadata including device make/model, capture timestamp, and embedded GPS coordinates is also extracted.

In person/keyword search mode, the system queries NewsAPI (1000+ global news sources), Reddit's public JSON API (all subreddits), and passes results to LLaMA 3.3 70B to generate a structured OSINT intelligence summary.

6.2 Cyber Intelligence

The Cyber Intelligence module provides five sub-modules: IP Tracker (geolocation, ISP identification, AbuseIPDB reputation scoring, VirusTotal multi-engine scanning, and AI threat assessment), Domain Investigator (WHOIS registration data, DNS record lookup, URLScan.io live scanning, VirusTotal domain reputation), Shodan Explorer (Internet-of-Things device discovery using structured search queries), Breach Checker (email breach lookup via LeakCheck.io), and VirusTotal Scanner (direct URL, IP, and domain submission to 70+ threat intelligence engines).

6.3 Geospatial Tracking

The Geo Tracker module integrates with OpenSky Network's OAuth2 API to provide real-time tracking of 10,000–14,000 aircraft globally, displayed on an interactive Leaflet.js map. Each aircraft marker shows ICAO code, callsign, country of registration, altitude, velocity, and heading. The module also provides geocoded location search with live weather data from OpenWeatherMap and links to Sentinel Hub satellite imagery and NASA Worldview.

6.4 News and Social Monitoring

The News and Social Monitor combines NewsAPI (100+ countries, 1000+ sources) with GDELT (the Global Database of Events, Language and Tone), which indexes media coverage in 65 languages from 150 countries. Sentiment analysis classifies each news query as POSITIVE, NEGATIVE, or NEUTRAL with a confidence score, feeding directly into the threat scoring model. LLaMA 3.3 70B generates an intelligence summary synthesising all retrieved articles.

6.5 Intelligence Graph

Inspired by Maltego-style link analysis tools, the Intelligence Graph module provides interactive network visualisation of entity relationships. For a given target entity, the system constructs a graph of nodes representing data sources (AbuseIPDB, VirusTotal, Shodan, WHOIS, LeakCheck, NewsAPI, etc.) and sub-nodes representing discovered intelligence. Investigators can zoom, pan, and click nodes to view details. The graph is rendered as an animated custom SVG using React, with no dependency on external graph libraries.

6.6 OSINT Playbooks

The Playbooks module provides six pre-built investigation templates: Person Investigation (6 steps), IP Threat Analysis (7 steps), Domain Reconnaissance (7 steps), Email Breach Investigation (6 steps), Image Forensics (6 steps), and Cyber Threat Hunt (7 steps). Steps animate sequentially and are directly linked to the relevant intelligence modules, enabling structured training in OSINT methodology.

7. Security Architecture

Security is a primary concern given the sensitive nature of intelligence operations. God's Eye implements multiple layers of security hardening.

7.1 Authentication and Authorisation

JWT-based authentication is implemented with role-based access control across three user tiers: ADMIN (full access including system logs and settings), RESEARCHER (all intelligence modules plus activity logs), and STUDENT (all intelligence modules, no administrative access). Tokens expire after 8 hours and embed the user's role in the payload, which is validated on every protected backend endpoint. CORS is restricted to localhost:5173 and localhost:5174, preventing cross-origin requests in production deployments.

7.2 API Key Protection

All 45+ API keys are stored exclusively in an .env file and loaded at backend startup via python-dotenv. Keys are never hardcoded in source code, never exposed to the frontend, and the .env file is listed in .gitignore to prevent accidental repository commits. The os.getenv().strip() pattern is used to handle Windows-specific line ending characters (\r) that can silently corrupt API keys loaded on that platform.

7.3 Activity Logging

All OSINT queries are logged with full accountability information: username, timestamp in IST, module used, action type, and target entity. This logging serves two purposes: it creates an audit trail for academic accountability, and it models real-world intelligence platform logging practices for educational value.

8. API Integration Catalogue

God's Eye integrates with 45+ external APIs across six functional domains. All APIs used are free-tier or publicly accessible, eliminating cost barriers for educational deployment. The following table summarises the primary integrations by category.

Category	API / Service	Key Capability
AI & Intelligence	Groq (LLaMA 3.3 70B)	Report generation, chat, summarisation
AI & Intelligence	Google Gemini	Backup AI inference
Face & Identity	Luxand.cloud	Age, gender, emotion, smile, glasses
Face & Identity	Face++	Facial attributes, beauty scoring
Face & Identity	SauceNAO	Reverse image search
Breach Detection	LeakCheck.io	Email breach database (Source 1)
Breach Detection	BreachDirectory	Email breach database (Source 2)
Breach Detection	HaveIBeenPwned	Industry-standard breach database
Cyber Threat	VirusTotal	70+ engine malware/URL scanner
Cyber Threat	AbuseIPDB	IP abuse reputation 0–100%
Cyber Threat	Shodan	Internet-connected device discovery
Cyber Threat	URLScan.io	Live website scanning
Geospatial	OpenSky Network	Live global flight tracking (OAuth2)
Geospatial	OpenWeatherMap	Live weather data
Geospatial	Sentinel Hub	Satellite imagery
News & Social	NewsAPI	1000+ global news sources
News & Social	GDELT Project	65 languages, 150 countries
News & Social	Reddit JSON API	Public subreddit monitoring

9. Results and System Performance

9.1 Parallel Processing Performance

The adoption of asyncio.gather() for parallel API dispatch produces significant performance improvements over sequential querying. A face scan involving Luxand, Face++, SauceNAO, and Yandex completes in approximately 5 seconds with parallel execution, compared to approximately 20 seconds if queries were dispatched sequentially. This 4× speed improvement is consistent across modules and is critical for interactive use in educational demonstrations.

9.2 Threat Score Validation

The Malicious IP preset (VirusTotal: 12 malicious engines, AbuseIPDB: 87%, domain age: 30 days, one breach) produces a composite score of 84/100 (CRITICAL), which aligns with the expected severity for a confirmed malicious IP address with active abuse reports. Clean infrastructure (e.g., 8.8.8.8 — Google's public DNS) produces a score of 0/100 (LOW) across all six signals, confirming that the model correctly identifies benign targets.

9.3 Live Data Capacity

The Geo Tracker module reliably retrieves 10,000–14,000 live aircraft positions from OpenSky Network at peak hours, demonstrating the platform's ability to handle and render large-scale real-time datasets. The dashboard's live threat feed updates every 2.5 seconds with no perceptible degradation in UI responsiveness.

10. Future Work and Development Roadmap

Several enhancements are planned for future releases of the God's Eye platform:

Priority	Feature	Description
HIGH	Docker containerisation	docker-compose deployment for Mac/Linux compatibility
HIGH	Deepfake detection	Open-source GAN artefact detection in uploaded images
HIGH	Mobile responsive design	Tailwind breakpoints for phone and tablet interfaces
MEDIUM	OSINT Alert system	Background keyword monitoring with push notifications
MEDIUM	Browser extension	Right-click any webpage for instant OSINT lookup
LOW	Multi-LLM support	User-selectable Groq, OpenAI, or local Ollama backend
LOW	Real-time collaboration	Shared investigation sessions for team-based OSINT
LOW	Cryptocurrency tracking	Bitcoin/Ethereum wallet address analysis

11. Ethical and Legal Framework

God's Eye is built exclusively on publicly available data sources and is designed for educational and authorised research use. The platform enforces ethical usage through several mechanisms: a mandatory Ethics Agreement checkbox at every login (no session begins without explicit consent), comprehensive activity logging for accountability, role-based access controls limiting capabilities to authorised users, and educational watermarks on all exported reports.

11.1 Permitted Uses

- Academic research and learning about OSINT and cybersecurity
- Investigating one's own digital footprint and privacy exposure
- Authorised security research with written permission from the target organisation
- Educational demonstrations and university portfolio projects
- Non-commercial cybersecurity training

11.2 Prohibited Uses

- Stalking, harassment, or tracking private individuals without consent
- Accessing private accounts, messages, or non-public data
- Bypassing any authentication or security system
- Using collected data for commercial purposes without licensing
- Any use that violates local, national, or international law

11.3 AI Ethics and Synthetic Media Awareness

In the context of the current proliferation of AI-generated synthetic media in 2025–2026, God's Eye includes a deepfake awareness notice on all exported intelligence reports. Users are instructed to cross-verify faces, quotes, and claims from multiple independent sources before drawing conclusions. A future high-priority feature is the integration of open-source GAN artefact detection to flag potentially synthetic images submitted for facial analysis.

12. Conclusion

This paper has presented God's Eye (OMNIVISION), a comprehensive educational OSINT platform integrating 45+ public APIs, asynchronous parallel data aggregation, AI-powered intelligence analysis, and a multi-signal threat scoring model. The system demonstrates that professional-grade intelligence capabilities can be delivered through a combination of free-tier APIs, modern full-stack web technologies, and large language model integration.

The platform contributes to the field of cybersecurity education by providing a structured, ethical, and accessible environment for students to develop practical OSINT skills. The standardised investigation playbooks, intelligence graph visualisation, and AI-generated reporting pipeline offer pedagogical value that exceeds what is available in existing open-source OSINT tools.

Future work will focus on cross-platform deployment via Docker, deepfake detection integration, and mobile-responsive design to broaden accessibility. The God's Eye project represents a meaningful step toward democratising intelligence analysis education while maintaining rigorous ethical standards.

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